

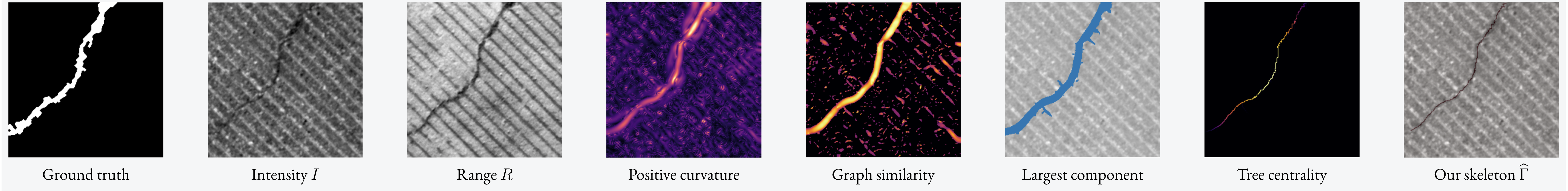
Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

Louis HAUSEUX¹, Raphaël ANTOINE², Philippe FOUCHER², Pierre CHARBONNIER², Josiane ZERUBIA¹

¹Inria, Université Côte d’Azur, Ayana team, France

²Cerema, END SUM research team, France

FIND first example. [1]

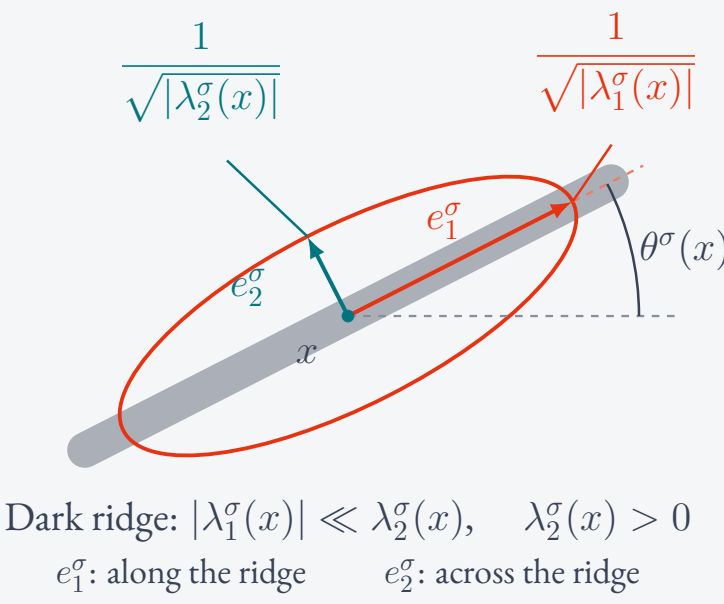


1. Fuse the modalities

Normalize each modality.

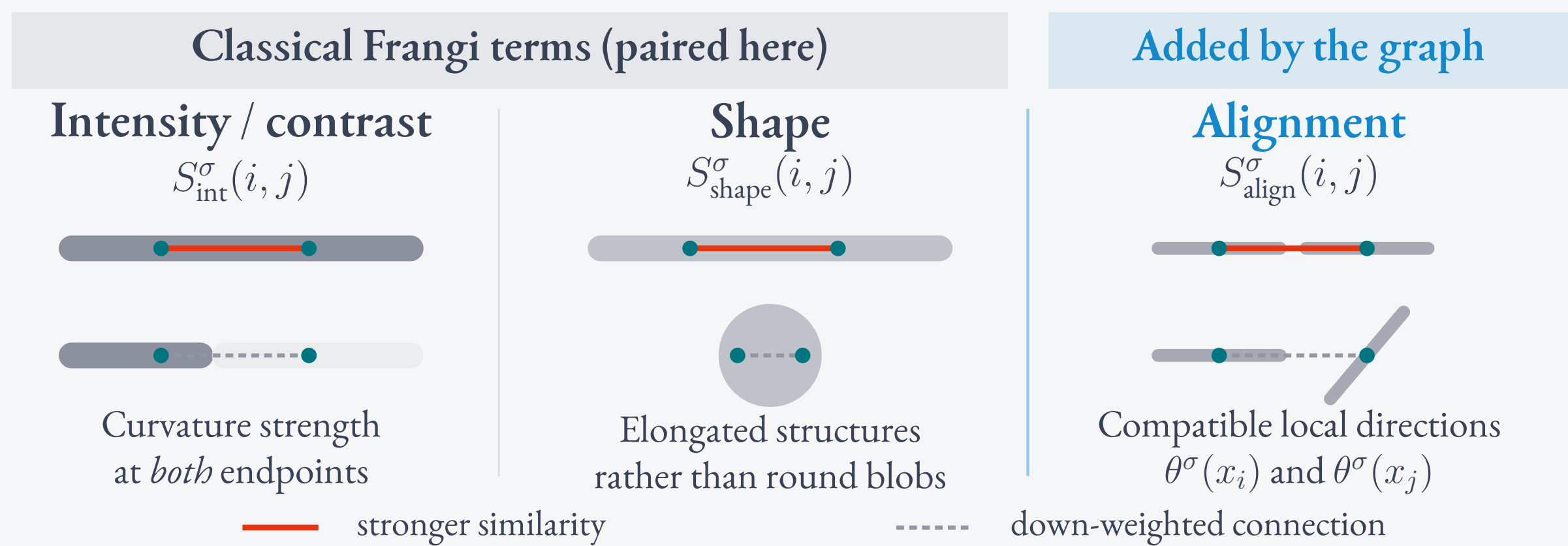
$$\hat{\mathcal{H}}_{\sigma}^{(m)}(x) = \frac{\mathcal{H}_{\sigma}^{(m)}(x)}{\max_{y \in \Omega} \left\| \mathcal{H}_{\sigma}^{(m)}(y) \right\|_2}.$$

Aligned dark-ridge inputs; $w_m \geq 0$, $\sum_m w_m = 1$.



Extract the geometry after fusion.
At scale σ : $|\lambda_1^{\sigma}(x)| \leq |\lambda_2^{\sigma}(x)|$.
Ridge direction: θ^{σ} .

2. From pixel-wise FRANGI [2] to pairwise links



Classical Frangi terms + neighbor alignment. For pixels with $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

$$C_{\sigma}(x) = \max\{\lambda_2^{\sigma}(x), 0\}, \quad R_B^{\sigma}(x) = \frac{|\lambda_1^{\sigma}(x)|}{|\lambda_2^{\sigma}(x)|}.$$

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^{\sigma}(i, j) S_{\text{shape}}^{\sigma}(i, j) S_{\text{align}}^{\sigma}(i, j)$$

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

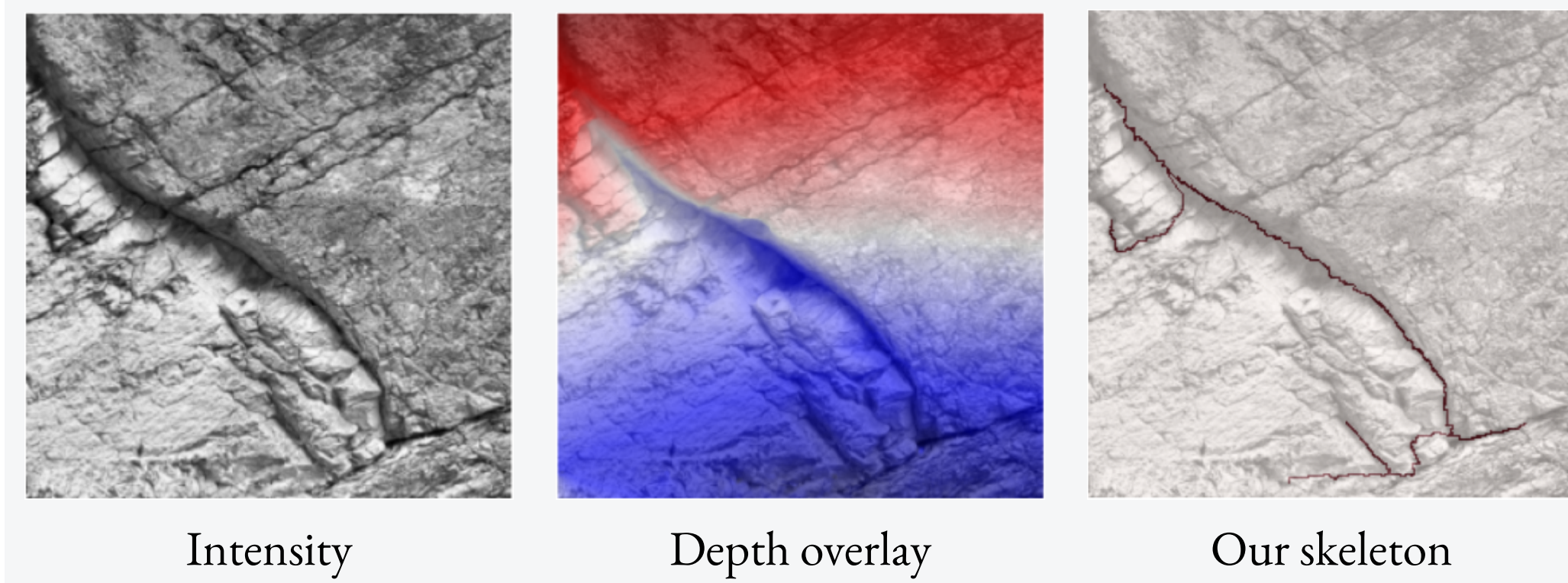
ρ_{ij} penalizes long links; θ and $\theta + \pi$ denote the same axis.

3. Extract the centerline



Per component: minimum spanning tree with costs d_{ij} , then similarity-weighted centrality. FIND settings: $\Sigma = \{1, 3, 5, 7\}$, $R = 3$, retained fractions $\tau_E = \tau_V = 0.25$.

Multimodality: Palais des Papes (France)



Fuse visible intensity and
photogrammetric depth.

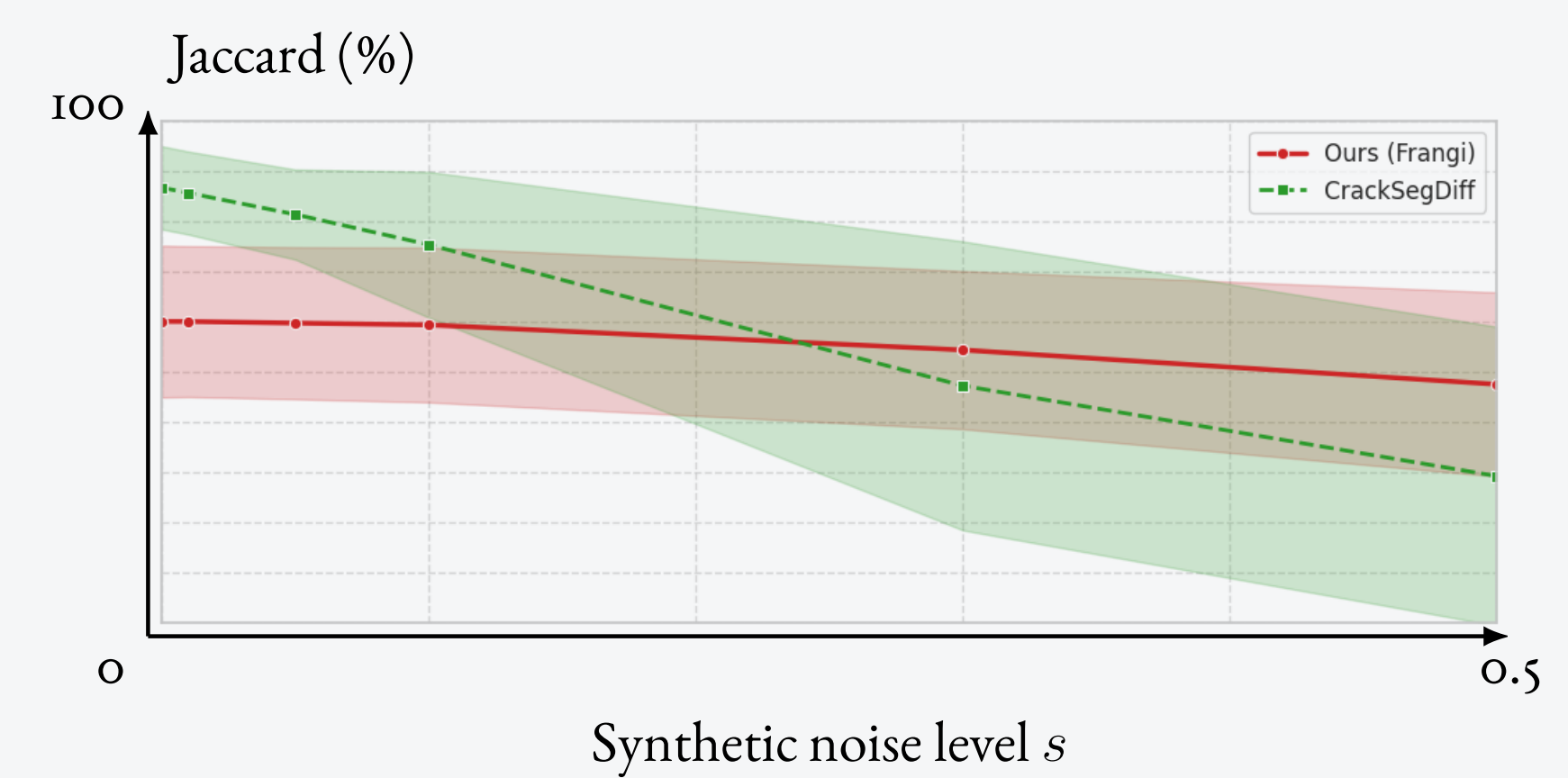
4. FIND: clean data

Method	Jaccard $\uparrow$	Tversky $\uparrow$	Wasserstein $\downarrow$
Ours, no training	63%	71%	11 px
CrackSegDiff [3]	87%	90%	5 px

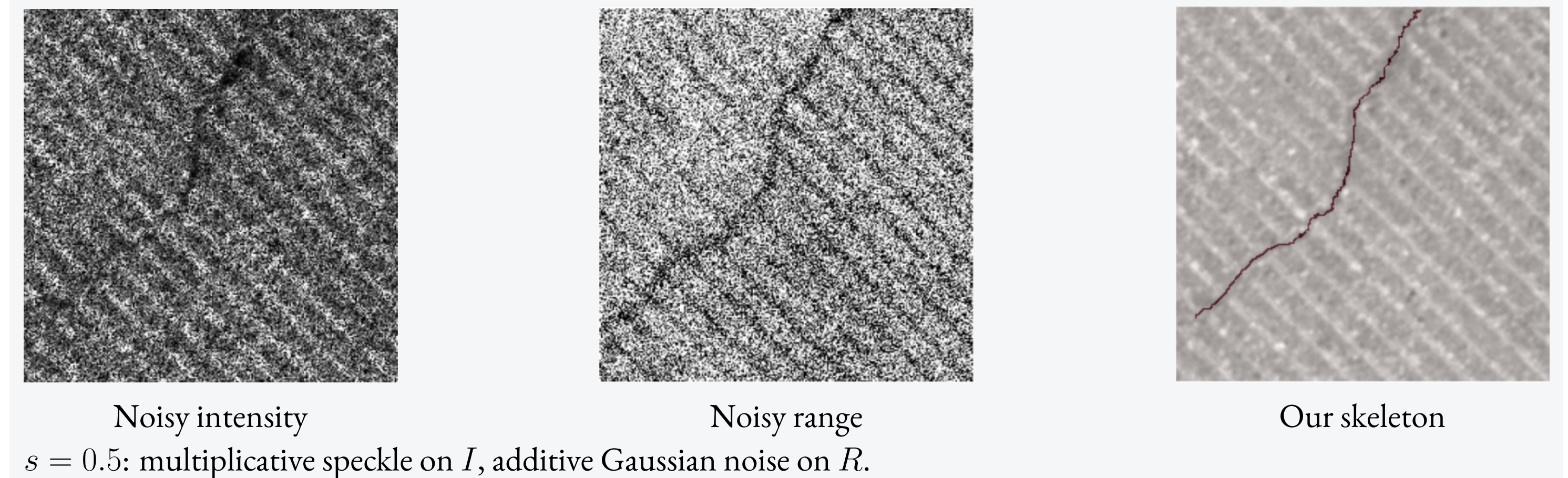
FIND: 256×256 . Overlap: both skeletons dilated by 3 px; Tversky: $\alpha = 1, \beta = 0.5$; Wasserstein: skeleton distance.

5. FIND: synthetic noise

Slower degradation under the tested synthetic noise.

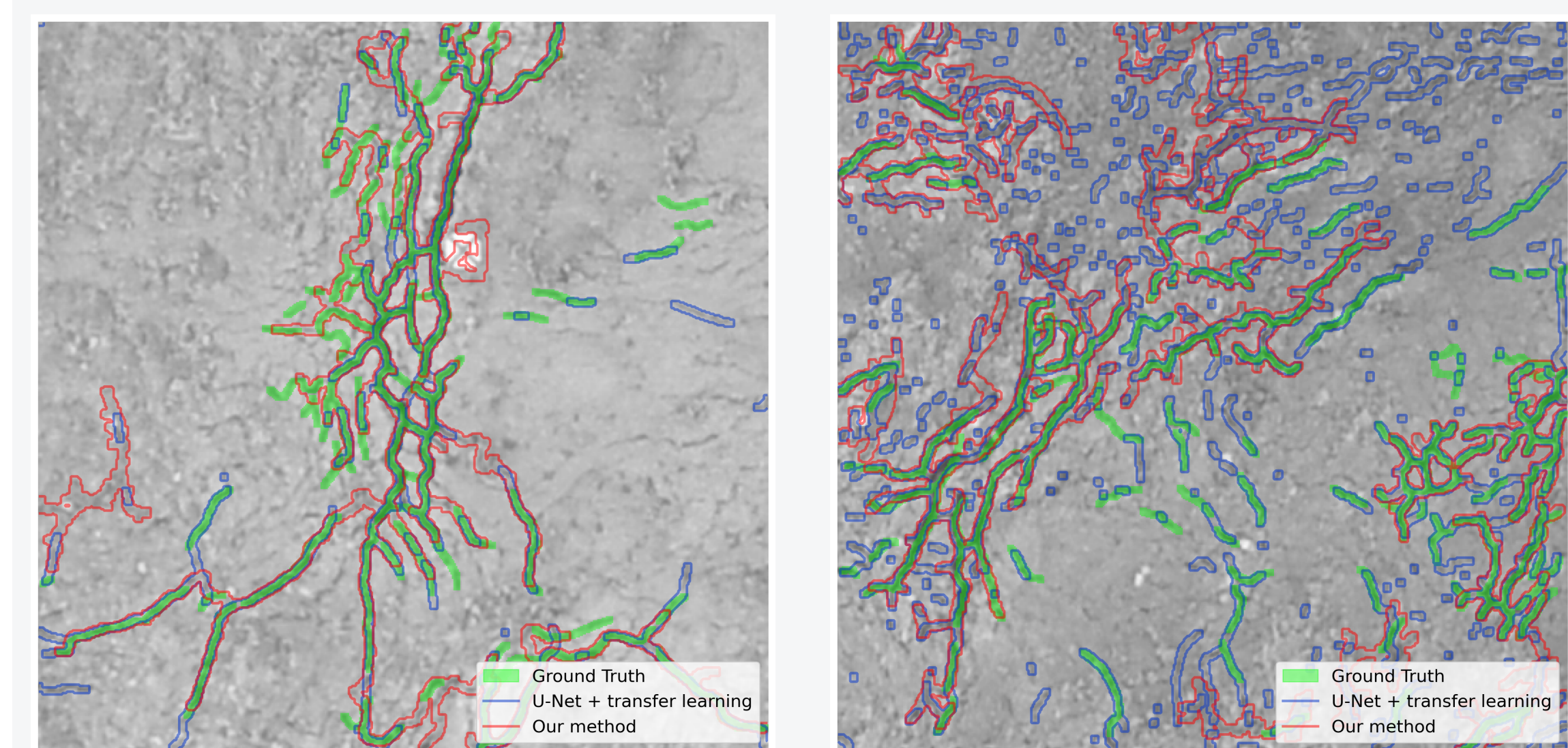


Bands: standard deviation.



6. Geological domain shift [4]

Vaches Noires. Two Cerema UAV patches (512×512).



Testr: 47% / 50%

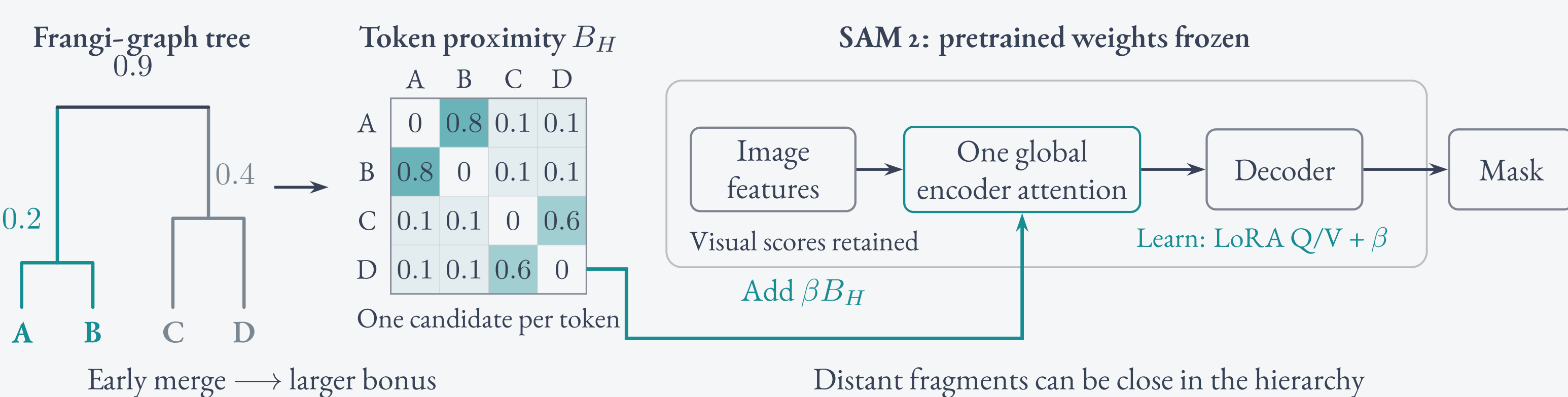
Test2: 29% / 34%

— Ours — U-Net + transfer learning — Ground truth

U-Net: about two days of expert annotation; ours: none.

Jaccard:
ours / U-Net.
Transfer strategy [5];
local U-Net trained on
a separate image.

Perspective: frozen SAM 2 + LoRA + Frangi hierarchy



SAM 2 [6]: frozen pretrained weights.

LoRA [7] + one coefficient β : learned together.

$$A_H = \text{softmax}(L + \beta B_H).$$

L : visual scores; B_H : hierarchy bias.

Merge heights describe connectivity [8]. We use them to bias attention, following Graphormer [9]. CrackSAM [10] motivates adaptation to cracks.

Aim: fewer breaks without more false links.

References

- [1] S. Zhou et al., Fused Image dataset for convolutional neural Network-based crack Detection (FIND). Zenodo, Data set, 2022.
- [2] A. F. Frangi et al., Multiscale vessel enhancement filtering. *MICCAI*, pp. 130–137, 1998.
- [3] X. Jiang et al., CrackSegDiff: Diffusion Probability Model-based Multi-modal Crack Segmentation. arXiv:2410.08100, 2024.
- [4] X. Zhang et al., Deep Learning for Crack Detection: A Review of Learning Paradigms, Generalizability, and Datasets. arXiv:2508.10256, 2025.
- [5] Y. Shi et al., Semi-universal geo-crack detection by machine learning. *Frontiers in Earth Science*, 11, 1073211, 2023.
- [6] N. Ravi et al., SAM 2: Segment Anything in Images and Videos. *ICLR*, 2025.
- [7] E. J. Hu et al., LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*, 2022.
- [8] S. C. Turaga et al., Maximin affinity learning of image segmentation. *NIPS*, 2009.
- [9] C. Ying et al., Do Transformers Really Perform Badly for Graph Representation? *NeurIPS*, 2021.
- [10] K. Ge et al., Fine-tuning vision foundation model for crack segmentation in civil infrastructures. *Construction and Building Materials*, 431, 136573, 2024.

louis.hauseux@inria.fr • github.com/Ayana-Inria/Frangi-EUVIP



Code & data